Limits

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Indeterminate forms: factor and cancel**

Explain why you cannot find $\lim_{x\to 4} \frac{x^2-16}{x-4}$ by direct substitution alone.

Substituting $x = 4$ gives $\frac{16-16}{4-4} = \frac{0}{0}$, an indeterminate form — it does not tell you the limit is 0 or undefined, so you must factor and cancel first.

2 **[Foundational]** • **Target: Limits at infinity for rational functions**

Explain what happens to $\lim_{x\to\infty} \frac{5x^3-2}{x^2+1}$, without computing an exact value.

The numerator's degree (3) is higher than the denominator's (2), so by dividing by the highest power in the denominator the numerator's leading term still grows without bound — the limit does not exist as a finite number; the function increases without bound as $x \to \infty$.

3 **[Foundational]** • **Target: Limits at infinity for rational functions**

Determine the slant asymptote of $g(x) = \frac{2x^3-3x^2+5}{x^2+1}$ and find the coordinates of any point where the curve intersects this asymptote.

Since $\deg(\text{num}) = 3$ and $\deg(\text{denom}) = 2$ with $3 - 2 = 1$, a slant asymptote exists. Perform polynomial long division: $(2x^3 - 3x^2 + 5) \div (x^2 + 1) = (2x - 3) + \frac{-2x+8}{x^2+1}$. The linear quotient gives the slant asymptote: $y = 2x - 3$. To find where the graph intersects this asymptote, set the remainder equal to zero: $\frac{-2x+8}{x^2+1} = 0 \Rightarrow -2x + 8 = 0 \Rightarrow x = 4$. Substitute $x = 4$ into the asymptote equation: $y = 2(4) - 3 = 5$. The curve intersects.

4 **[Foundational]** • **Target: Direct substitution**

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

5 **[Foundational]** • **Target: Indeterminate forms: factor and cancel**

Solve the system using matrix inversion: $4x + 7y = 11\ x + 2y = 3$

Step 1: Write as matrix equation $A\mathbf{x} = \mathbf{b}$: $(4, 7; 1, 2)(x; y) = (11; 3)$ Step 2: Multiply by $A^{-1} = (2, -7; -1, 4)$ (from Problem 14): $(x; y) = (2, -7; -1, 4)(11; 3) = (2(11) + (-7)(3); -1(11) + 4(3)) = (22 - 21; -11 + 12) = (\mathbf{1; 1})$. Thus $\mathbf{x = 1}$ and $\mathbf{y = 1}$.

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6 [Foundational] • Target: Limits at infinity for rational functions

Find all real solutions to the radical equation: $\sqrt{2x+15}-x=0$

Step 1: Isolate radical and square both sides: $\sqrt{2x+15}=x \Rightarrow 2x+15=x^2 \Rightarrow x^2-2x-15=0$ Step 2: Factor: $(x-5)(x+3)=0 \Rightarrow x=5 \quad \text{or} \quad x=-3$. Step 3: Check in original equation: For $x=5$: $\sqrt{10+15}-5=\sqrt{25}-5=0$ ✓. For $x=-3$: $\sqrt{-6+15}-(-3)=\sqrt{9}+3=3+3=6eq0$ (extraneous). Thus the unique solution is **x = 5**.

7 [Foundational] • Target: Direct substitution

In a geometric sequence, $a_1=5$ and $a_4=40$. Find the common ratio r and the 7th term a_7.

Step 1: Find r: $a_4=a_1r^3 \Rightarrow 40=5r^3 \Rightarrow r^3=8 \Rightarrow \mathbf{r=2}$. Step 2: Find a_7: $a_7=a_1r^6=5(2^6)=5(64)=\mathbf{320}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Indeterminate forms: factor and cancel

Calculate the determinant of the matrix $M=(7,3;\ -2,5)$.

Formula: For $(a,b;c,d)$, $\det(M)=ad-bc$. $\det(M)=(7)(5)-(3)(-2)=35-(-6)=35+6=\mathbf{41}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

9 [Intermediate] • Target: Limits at infinity for rational functions

Convert the repeating decimal $0.\overline{45}$ to an irreducible common fraction.

Step 1: Express as an infinite geometric series: $0.454545\ldots=\frac{45}{100}+\frac{45}{10000}+\cdots$ where $a_1=\frac{45}{100}$ and $r=\frac{1}{100}$. Step 2: Sum the series: $S_\infty=\frac{\frac{45}{100}}{1-\frac{1}{100}}=\frac{\frac{45}{100}}{\frac{99}{100}}=\frac{45}{99}=\mathbf{\frac{5}{11}}$.

10 [Intermediate] • Target: Direct substitution

Find the 25th term of the arithmetic sequence: $7,12,17,22,\ \ldots$

Step 1: Identify a_1 and d: $a_1=7$, $d=12-7=5$. Step 2: Apply the arithmetic formula: $a_n=a_1+(n-1)d$ $a_{25}=7+(25-1)(5)=7+24(5)=7+120=\mathbf{127}$.

11 [Intermediate] • Target: Indeterminate forms: factor and cancel

In triangle ABC, sides are $a=5$, $b=6$, and $c=7$. Find the exact value of $\cos A$.

Law of Cosines solved for $\cos A$: $\cos A=\frac{b^2+c^2-a^2}{2bc}$ $\cos A=\frac{6^2+7^2-5^2}{2(6)(7)}=\frac{36+49-25}{84}=\frac{60}{84}=\mathbf{\frac{5}{7}}$.

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12 **[Intermediate]** • **Target: Limits at infinity for rational functions**

If $z_1 = 5 - 2i$ and $z_2 = -3 + 7i$, find the value of $2z_1 - z_2$ in standard form $a + bi$.

Step 1: Distribute scalar multiplication: $2z_1 = 2(5 - 2i) = 10 - 4i$ Step 2: Subtract z_2:

$2z_1 - z_2 = (10 - 4i) - (-3 + 7i) = (10 - (-3)) + (-4 - 7)i = \mathbf{13 - 11i}$.

13 **[Intermediate]** • **Target: Direct substitution**

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.

Step 1: Domain and vertical asymptotes: The denominator factors as $x^2 - 4 = (x - 2)(x + 2)$. The numerator $3x^2 + 1 > 0$ has no real zeros. Thus, there are no common factors and no removable holes. The vertical asymptotes are $x = 2$ and $x = -2$. Step 2: Horizontal asymptote and end behavior: Numerator and denominator have equal degree $n = d = 2$: $\lim_{x \to \pm\infty} f(x) = \lim_{x \to \pm\infty} \frac{3 + 1/x^2}{1 - 4/x^2} = \frac{3}{1} = 3$. The horizontal asymptote is $y = 3$. Notice tha.

14 **[Intermediate]** • **Target: Indeterminate forms: factor and cancel**

Find all real solutions to the rational equation $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2 + x - 6}$.

Notice that the common denominator factors as $x^2 + x - 6 = (x - 2)(x + 3)$. Domain restrictions: $x \neq 2$ and $x \neq -3$. Multiply both sides by $(x - 2)(x + 3)$: $x(x + 3) + 1(x - 2) = 5 \Rightarrow x^2 + 3x + x - 2 = 5$ $x^2 + 4x - 7 = 0$ Apply the quadratic formula with $a = 1, b = 4, c = -7$:

$x = \frac{-4 \pm \sqrt{4^2 - 4(1)(-7)}}{2(1)} = \frac{-4 \pm \sqrt{16 + 28}}{2} = \frac{-4 \pm \sqrt{44}}{2} = \frac{-4 \pm 2\sqrt{11}}{2} = \mathbf{-2 \pm \sqrt{11}}$ Neither solution equals 2 or -3, so both solutio.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Limits at infinity for rational functions** • **Cross-Lesson: 5.01 Trigonometry & Polar Coordinates**

Decompose the rational expression $R(x) = \frac{7x - 1}{x^2 - x - 6}$ into partial fractions using the Heaviside cover-up method.

Step 1: Factor denominator: $x^2 - x - 6 = (x - 3)(x + 2)$. Set up the decomposition: $\frac{7x - 1}{(x - 3)(x + 2)} = \frac{A}{x - 3} + \frac{B}{x + 2}$. Step 2: Apply Heaviside cover-up method:For A (evaluate at $x = 3$): $A = \frac{7x - 1}{x + 2}\Big|_{x = 3} = \frac{7(3) - 1}{3 + 2} = \frac{20}{5} = 4$. For B (evaluate at $x = -2$): $B = \frac{7x - 1}{x - 3}\Big|_{x = -2} = \frac{7(-2) - 1}{-2 - 3} = \frac{-15}{-5} = 3$. The partial fraction decomposition is $\frac{4}{x - 3} + \frac{3}{x + 2}$.

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16 **[Challenging]** • **Target: Direct substitution** • **Cross-Lesson: 1.01 Complex Numbers**

Let $f(x) = \frac{x^2 - 9}{x^2 - 5x + 6}$. Identify the locations of all vertical asymptotes and removable discontinuities (holes) of $f(x)$.

Factor numerator and denominator completely: $f(x) = \frac{(x-3)(x+3)}{(x-2)(x-3)}$ Notice that $(x-3)$ is a common factor in both numerator and denominator: Removable Discontinuity (Hole): Occurs at $x = 3$. To find the y -coordinate, evaluate the simplified fraction: $f_{\text{reduced}}(3) = \frac{3+3}{3-2} = \frac{6}{1} = 6$. The hole is at **(3, 6)**.Vertical Asymptote: The remaining uncancelled denominator factor is $x - 2 = 0$, creating a vertical asymptote at line $\mathbf{x = 2}$.

17 **[Challenging]** • **Target: Indeterminate forms: factor and cancel** • **Cross-Lesson: 2.01 Matrices & Vectors**

Convert the repeating decimal $x = 0.4\overline{36} = 0.4363636\ldots$ into an irreducible rational fraction $\frac{p}{q}$.

Let $x = 0.4\overline{36}$. Multiply by powers of 10 to shift the non-repeating and repeating parts: $10x = 4.\overline{36}$ $1000x = 436.\overline{36}$ Subtract the first equation from the second to eliminate the repeating tails: $1000x - 10x = 436.\overline{36} - 4.\overline{36} \Rightarrow 990x = 432$ $x = \frac{432}{990}$ Divide numerator and denominator by their greatest common factor (18): $432 \div 18 = 24, \quad 990 \div 18 = 55 \Rightarrow x = \mathbf{\frac{24}{55}}$

18 **[Challenging]** • **Target: Limits at infinity for rational functions** • **Cross-Lesson: 3.01 Sequences & Series**

Determine whether the function $g(x) = \frac{x^3}{x^2 + 4}$ is even, odd, or neither.

Evaluate $g(-x)$ by substituting $-x$ for every occurrence of x: $g(-x) = \frac{(-x)^3}{(-x)^2 + 4} = \frac{-x^3}{x^2 + 4} = -\left(\frac{x^3}{x^2 + 4}\right) = -g(x)$ Because $g(-x) = -g(x)$ for all $x \in \mathbb{R}$, $g(x)$ is an odd function (symmetric with respect to 180° rotation about the origin).

19 **[Challenging]** • **Target: Direct substitution** • **Cross-Lesson: 4.01 Advanced Functions & Equations**

If $f(x) = 2x - 5$ and $g(x) = x^2 + 3$, find the expression for $(f \circ g)(x)$ and solve for all x such that $(f \circ g)(x) = 25$.

Step 1: Form the composite function $(f \circ g)(x) = f(g(x))$: $f(g(x)) = f(x^2 + 3) = 2(x^2 + 3) - 5 = 2x^2 + 6 - 5 = \mathbf{2x^2 + 1}$ Step 2: Solve $2x^2 + 1 = 25$: $2x^2 = 24 \Rightarrow x^2 = 12 \Rightarrow x = \pm\sqrt{12} = \mathbf{\pm 2\sqrt{3}}$

20 **[Challenging]** • **Target: Indeterminate forms: factor and cancel** • **Cross-Lesson: 5.01 Trigonometry & Polar Coordinates**

A function is defined piecewise: $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$. What is the value of $f(-3) + f(2)$?

Evaluate each term according to its interval condition: For $x = -3 < 0$: $f(-3) = (-3)^2 = 9$. For $x = 2 \geq 0$: $f(2) = 2(2) + 1 = 5$. Sum: $f(-3) + f(2) = 9 + 5 = \mathbf{14}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Direct substitution

Find $\lim_{x\to\infty} \frac{x^3+5x}{9x^2-9x}$ (state whether it is ∞ or $-\infty$).

a. $\frac{1}{9}$

b. 0

c. $-\infty$

d. ∞ ✓

The numerator's degree (3) is higher than the denominator's (2), so the quotient behaves like $\frac{1}{9}x$ for large x. Since the leading coefficients 1 and 9 have same sign, the limit is ∞.

22 [Foundational] • Target: Indeterminate forms: factor and cancel

For real numbers a and b, suppose that $\lim_{x\to1} \frac{x^2+ax+b}{x^2-1} = 3$. What is the value of $a-b$?

a. -9

b. -1

c. 3

d. **9** ✓

For the rational function to have a finite limit as $x \to 1$ while the denominator $x^2-1 \to 0$, the numerator must also approach 0. Setting the numerator to 0 at $x=1$:
$1^2+a(1)+b=0 \Rightarrow 1+a+b=0 \Rightarrow b=-(a+1)$. Substitute $b=-(a+1)$ back into the numerator:
$x^2+ax-(a+1)=(x-1)(x+a+1)$. Now evaluate the limit by canceling the common factor $(x-1)$.

23 [Foundational] • Target: Limits at infinity for rational functions

What is the value of the limit $\lim_{x\to2} \frac{x^2+x-6}{x-2}$?

a. 5 ✓

b. 1

c. -1

d. Does not exist

Direct substitution of $x=2$ gives the indeterminate form: $\frac{2^2+2-6}{2-2}=\frac{0}{0}$ To evaluate the limit, factor the quadratic expression in the numerator: $x^2+x-6=(x-2)(x+3)$ For all $xeq2$, we can cancel the common factor $(x-2)$: $\frac{x^2+x-6}{x-2}=\frac{(x-2)(x+3)}{x-2}=x+3$ Now, evaluate the limit by direct substitution: $\lim_{x\to2}(x+3)=2+3=5$ Common misconceptions include: - Factoring incorrectly as $(x-2)(x-3)$, leading to.

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24 [Foundational] • Target: Direct substitution

For $f(x) = 2x^3 + 2x^2 - x + 6$, which values are possible rational roots (by the Rational Root Theorem)?

a. $\pm 1, \pm 2$

b. ± 6

c. ± 12

d. $\pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}, \pm 2, \pm 3, \pm 6$ ✓

Possible rational roots are $\pm\frac{p}{q}$ where p divides the constant term 6 and q divides the leading coefficient 2: $\pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}, \pm 2, \pm 3, \pm 6$.

25 [Foundational] • Target: Indeterminate forms: factor and cancel

Find all vertical asymptotes of $f(x) = \frac{x^2 + x - 6}{x^3 - 6x^2 + 3x + 10}$.

a. x=2,-1,5

b. x=2

c. x=-1

d. **x=-1,5** ✓

Factor: $f(x) = \frac{(x-2)(x+3)}{(x-2)(x-5)(x+1)}$. The $(x-2)$ factor cancels (hole at $x = 2$), leaving vertical asymptotes at $x = -1, 5$.

26 [Foundational] • Target: Limits at infinity for rational functions

For $f(x) = \frac{(x+2)(x+1)}{(x+1)(x+3)}$, identify the hole and vertical asymptote.

a. Hole at $x = -3$, vertical asymptote at $x = -1$

b. **Hole at $x = -1$, vertical asymptote at $x = -3$** ✓

c. Vertical asymptotes at $x = -1$ and $x = -3$, no hole

d. Hole at $x = -2$, vertical asymptote at $x = -3$

The factor $(x + 1)$ cancels, creating a hole at $x = -1$; the remaining zero of the denominator, $x = -3$, gives the vertical asymptote.

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27 [Foundational] • Target: Direct substitution

For $f(x) = (x - 4)^3(x + 10)$, describe the behavior of the graph at $x = 4$.

a. crosses the x-axis normally (linear)

b. **crosses the x-axis, flattening (S-shaped) near the zero ✓**

c. touches the x-axis and turns around (does not cross)

d. the graph has a vertical asymptote there

The zero $x = 4$ has multiplicity 3, which is odd, so the graph crosses the x-axis, flattening (S-shaped) near the zero.

28 [Foundational] • Target: Indeterminate forms: factor and cancel

As $x \to +\infty$, what happens to $f(x) = 2x^5$?

a. $+\infty$ ✓

b. $-\infty$

c. 0

d. Oscillates

Leading coefficient is 2 (positive), degree 5. As $x \to +\infty$, $f(x) \to +\infty$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Limits at infinity for rational functions

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x + 1}{x - 1}$.

a. $y = 6x$

b. $y = x + 6$ ✓

c. $y = x + 7$

d. No slant asymptote

Divide: $x^2 + 5x + 1 = (x - 1)(x + 6) + 7$. Dropping the remainder term, the slant asymptote is $y = x + 6$.

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30 [Intermediate] • Target: Direct substitution

Find the x-coordinate of the hole in the graph of $f(x) = \frac{x^2 + x - 6}{x^2 - 7x + 10}$.

a. x=5

b. **x=2 ✓**

c. No hole exists

d. x=-3

$f(x) = \frac{(x-2)(x+3)}{(x-2)(x-5)}$; the common factor $(x - 2)$ gives a hole at $x = 2$.

31 [Intermediate] • Target: Indeterminate forms: factor and cancel

Find the horizontal asymptote of $f(x) = \frac{x^3 + 1}{x^2 - 4}$, or state that none exists.

a. The line $y = x$

b. y=0

c. y=1

d. No horizontal asymptote ✓

Degree of numerator $>$ degree of denominator $\Rightarrow$ no horizontal asymptote.

32 [Intermediate] • Target: Limits at infinity for rational functions

Using the Remainder Theorem, find the remainder when $f(x) = 2x^2 - 1x + 6$ is divided by $x + 3$.

a. -27

b. 28

c. **27 ✓**

d. 26

By the Remainder Theorem, the remainder equals $f(-3) = 2(-3)^2 - 1(-3) + 6 = 27$.

33 [Intermediate] • Target: Direct substitution

What is the horizontal asymptote of $f(x) = \frac{1}{x}$?

a. $y = 1$

b. $y = 0$ ✓

c. None

d. $x = 0$

Degree of numerator (0) is less than degree of denominator (1), so $y = 0$.

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34 [Intermediate] • Target: Indeterminate forms: factor and cancel

What is the vertical asymptote of $f(x) = \frac{1}{x-0}$?

a. $x = 0$ ✓

b. $y = 0$

c. $x = 1$

d. None

The denominator is zero at $x = 0$, giving a vertical asymptote there.

35 [Intermediate] • Target: Limits at infinity for rational functions

Find the horizontal asymptote of $f(x) = \frac{3}{x^2-4}$.

a. **y=0** ✓

b. y=-1

c. y=1

d. y=3

Degree of numerator $<$ degree of denominator $\Rightarrow y = 0$.

36 [Intermediate] • Target: Direct substitution

Simplify $f(x) = \frac{x^2+x-6}{x^2-7x+10}$ and evaluate at $x = 1$.

a. 1

b. 0

c. 2

d. **-1** ✓

$f(x) = \frac{x+3}{x-5}$ for $xeq2$; at $x = 1$: $\frac{4}{-4} = -1$.

37 [Intermediate] • Target: Indeterminate forms: factor and cancel

What is the degree and leading coefficient of $f(x) = 2x^4 + x^2 - 2$?

a. Degree 5, leading coefficient 2

b. **Degree 4, leading coefficient 2** ✓

c. Degree 4, leading coefficient -2

d. Degree 3, leading coefficient 2

The highest power of x is 4, with coefficient 2.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Limits at infinity for rational functions • Cross-Lesson: 3.01 Sequences & Series

Find the vertical asymptote of $f(x) = \frac{2}{x-3}$.

a. y=3

b. x=0

c. **x=3 ✓**

d. x=-3

Denominator $= 0 \Rightarrow x - 3 = 0 \Rightarrow x = 3$.

39 [Challenging] • Target: Direct substitution • Cross-Lesson: 4.01 Advanced Functions & Equations

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.

a. **0 ✓**

b. -2

c. 2

d. 1

The general form of the quadratic function is $y = ax^2 + bx + c$. We are given three points that lie on the graph: $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Substituting these points into the equation, we get a system of three linear equations with three unknowns (a, b, and c): 1. For $(-1, 5)$: $a(-1)^2 + b(-1) + c = 5 \Rightarrow a - b + c = 5$ 2. For $(1, 1)$: $a(1)^2 + b(1) + c = 1 \Rightarrow a + b + c = 1$ 3. For $(3, 5)$: $a(3)^2 + b(3) + c = 5 \Rightarrow 9a + 3b + c = 5$ We can solve this system of equations. Le.

40 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$ **✓**

c. 1

d. $\frac{\sqrt{2}}{2}$

The equation of a unit circle centered at the origin is $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $y = \frac{\sqrt{3}}{2}$. Substituting this into the equation of the unit circle: $x^2 + (\frac{\sqrt{3}}{2})^2 = 1$ $x^2 + \frac{3}{4} = 1$ $x^2 = 1 - \frac{3}{4}$ $x^2 = \frac{1}{4}$ Since point P lies in the first quadrant, its x-coordinate must be positive. Therefore, $x = \sqrt{\frac{1}{4}} = \frac{1}{2}$. Alternatively, we are given that the angle θ formed by the ray from the or.

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41 [Challenging] • Target: Limits at infinity for rational functions • Cross-Lesson: 1.01 Complex Numbers

Consider the functions $f(x) = x^2 - 6x + 5$ and $g(x) = -x + 5$. The graphs of these two functions intersect at two points. Let these points be A and B. A third function, $h(x)$, is defined as the difference between the x-coordinates of these intersection points, i.e., $h(x) = |x_A - x_B|$. What is the value of $h(x)$?

a. $\sqrt{16}$

b. $\sqrt{25}$ ✓

c. $\sqrt{36}$

d. $\sqrt{49}$

To find the intersection points of the graphs of $f(x)$ and $g(x)$, we set $f(x) = g(x)$. $x^2 - 6x + 5 = -x + 5$ Subtract $-x + 5$ from both sides: $x^2 - 6x + 5 - (-x + 5) = 0$ $x^2 - 6x + 5 + x - 5 = 0$ Combine like terms: $x^2 - 5x = 0$ Factor out x: $x(x - 5) = 0$ This gives two possible x-values for the intersection points: $x_1 = 0$ $x_2 = 5$ Let $x_A = 0$ and $x_B = 5$ (or vice versa). The problem defines $h(x)$ as the absolute difference between these x-coordinates: $h(x) = |x_A - x_B|$..

42 [Challenging] • Target: Direct substitution • Cross-Lesson: 2.01 Matrices & Vectors

The graph of the quadratic function $f(x) = x^2 + bx + c$ is shown below. The graph passes through the point $(-2, 5)$ and has its vertex at $(1, -4)$. Determine the value of b.

a. 1

b. **-2** ✓

c. -1

d. 2

The vertex of a parabola given by $f(x) = ax^2 + bx + c$ is at $x = -\frac{b}{2a}$. In this case, $a = 1$ and the x-coordinate of the vertex is given as 1. So, $1 = -\frac{b}{2(1)}$. This simplifies to $1 = -\frac{b}{2}$, which means $b = -2$. Alternatively, we can use the vertex form of a quadratic equation: $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Here, $a = 1$ and $(h, k) = (1, -4)$, so $f(x) = 1(x - 1)^2 - 4$. Expanding this, we get $f(x) = (x^2 - 2x + 1) - 4 = x^2 - 2x - 3$. Comparing this to.

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43 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 3.01 Sequences & Series

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(2, -5)$ and the y-intercept is at $(0, -1)$. Determine the value of a.

a. **1** ✓

b. -1

c. 2

d. -2

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(2, -5)$, the function can be written as $f(x) = a(x - 2)^2 - 5$. We are also given that the y-intercept is $(0, -1)$. This means that when $x = 0$, $f(x) = -1$. Substituting these values into the vertex form: $-1 = a(0 - 2)^2 - 5$. This simplifies to $-1 = a(-2)^2 - 5$, which is $-1 = 4a - 5$. Adding 5 to both sides gives $4 = 4a$. Dividing.

44 [Challenging] • Target: Limits at infinity for rational functions • Cross-Lesson: 4.01 Advanced Functions & Equations

Consider the function $f(x) = x^2 - 4x + 3$. What are the coordinates of the vertex of the parabola represented by this function?

a. **(2, -1)** ✓

b. (2, 1)

c. (1, 0)

d. (3, 0)

The given function is a quadratic function in the form $f(x) = ax^2 + bx + c$, where $a = 1$, $b = -4$, and $c = 3$. The x-coordinate of the vertex of a parabola is given by the formula $x = -\frac{b}{2a}$. Substituting the values, we get $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. To find the y-coordinate of the vertex, we substitute this x-value back into the function: $f(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Therefore, the coordinates of the vertex are (2, -1).

45 [Challenging] • Target: Direct substitution • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Determine the y-coordinate of the vertex of the parabola.

a. **1** ✓

b. 2

c. 3

d. 4

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. In this case, $a = -1$ and $b = 4$. So, the x-coordinate of the vertex is $x = -\frac{4}{2(-1)} = -\frac{4}{-2} = 2$. To find the y-coordinate of the vertex, substitute this x-value back into the function: $f(2) = -(2)^2 + 4(2) - 3 = -4 + 8 - 3 = 1$. Therefore, the vertex is at (2, 1) and the y-coordinate is 1.

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